

Clinical management and therapeutic optimization of patients with heart failure with reduced ejection fraction and low blood pressure. A clinical consensus statement of the Heart Failure Association (HFA) of the ESC

Hadi Skouri^{1,2*}, Nicolas Girerd^{3†}, Luca Monzo³, Mark C. Petrie⁴, Michael Böhm⁵, Marianna Adamo⁶, Wilfried Mullens^{7,8}, Gianluigi Savarese⁹, Mehmet Birhan Yilmaz¹⁰, Offer Amir¹¹, Antoni Bayes-Genis¹², Biykem Bozkurt¹³, Javed Butler^{14,15}, Ovidiu Chioncel¹⁶, Alexandre Mebazaa¹⁷, Jose L. Merino¹⁸, Brenda Moura¹⁹, Piotr Ponikowski²⁰, Petar Seferovic²¹, Giuseppe M.C. Rosano^{22,23,24,25}, and Marco Metra⁶

¹Division of Cardiology, Balamand University Faculty of Medicine, Beirut, Lebanon; ²Division of Cardiology, Sheikh Shakhboub Medical City, Abu-Dhabi, UAE; ³Université de Lorraine, Centre d'Investigations Cliniques Plurithématique 1433 and Inserm U1116, CHRU Nancy, FCRIN INI-CRCT (Cardiovascular and Renal Clinical Trialists), Nancy, France; ⁴School of Cardiovascular and Metabolic Health, University of Glasgow, British Heart Foundation Glasgow Cardiovascular Research Centre, Glasgow, UK; ⁵Universitätsklinikum des Saarlandes, Klinik für Innere Medizin III, Saarland University, Kardiologie, Angiologie und Internistische Intensivmedizin, Homburg, Germany; ⁶Cardiology and Cardiac Catheterization Laboratory, ASST Spedali Civili Brescia and Department of Medical and Surgical Specialties, Radiological Sciences, and Public Health, University of Brescia, Brescia, Italy; ⁷Department of Cardiology, Ziekenhuis Oost-Limburg, Genk, Belgium; ⁸U Hasselt, Biomedical Research Institute, Faculty of Medicine and Life Sciences, LCRC, Diepenbeek, Belgium; ⁹Unit of Cardiology, Department of Medicine, Karolinska Institutet, Stockholm, Sweden; ¹⁰Department of Cardiology, Faculty of Medicine, Dokuz Eylül University, İzmir, Türkiye; ¹¹Hadassah Medical Center, Faculty of Medicine, Hebrew University, Jerusalem, Israel; ¹²Heart Institute, Hospital Universitari Germans Trias i Pujol, CIBERCV, Badalona, Spain; ¹³Cardiology Section, Winters Center for Heart Failure, Baylor College of Medicine, Houston, TX, USA; ¹⁴Baylor Scott and White Research Institute, Dallas, TX, USA; ¹⁵University of Mississippi, Jackson, MS, USA; ¹⁶Emergency Institute for Cardiovascular Diseases 'Prof. C.C. Iliescu', University of Medicine Carol Davila, Bucharest, Romania; ¹⁷Department of Anaesthesia-Burn-Critical Care, Université de Paris, UMR Inserm – MASCOT; APHP Saint Louis Lariboisière University Hospitals, Paris, France; ¹⁸Arrhythmia & Robotic EP Unit, University Hospital La Paz, Autonomia University, IdiPaz, Madrid, Spain; ¹⁹Armed Forces Hospital, Porto, and Faculty of Medicine of Porto, Porto, Portugal; ²⁰Department of Heart Diseases, Wrocław Medical University, Wrocław, Poland; ²¹University Faculty of Medicine, University of Belgrade, and Serbian Academy of Arts and Sciences, Belgrade, Serbia; ²²Department of Human Sciences and Promotion of Quality of Life, San Raffaele Open University of Rome, Rome, Italy; ²³Cardiology, San Raffaele Cassino Hospital, Cassino, Italy; ²⁴IRCCS San Raffaele Roma, Roma, Italy; and ²⁵Cardiology Clinical Academic Group, Molecular and Clinical Sciences Research Institute, City St George's, University of London, London, UK

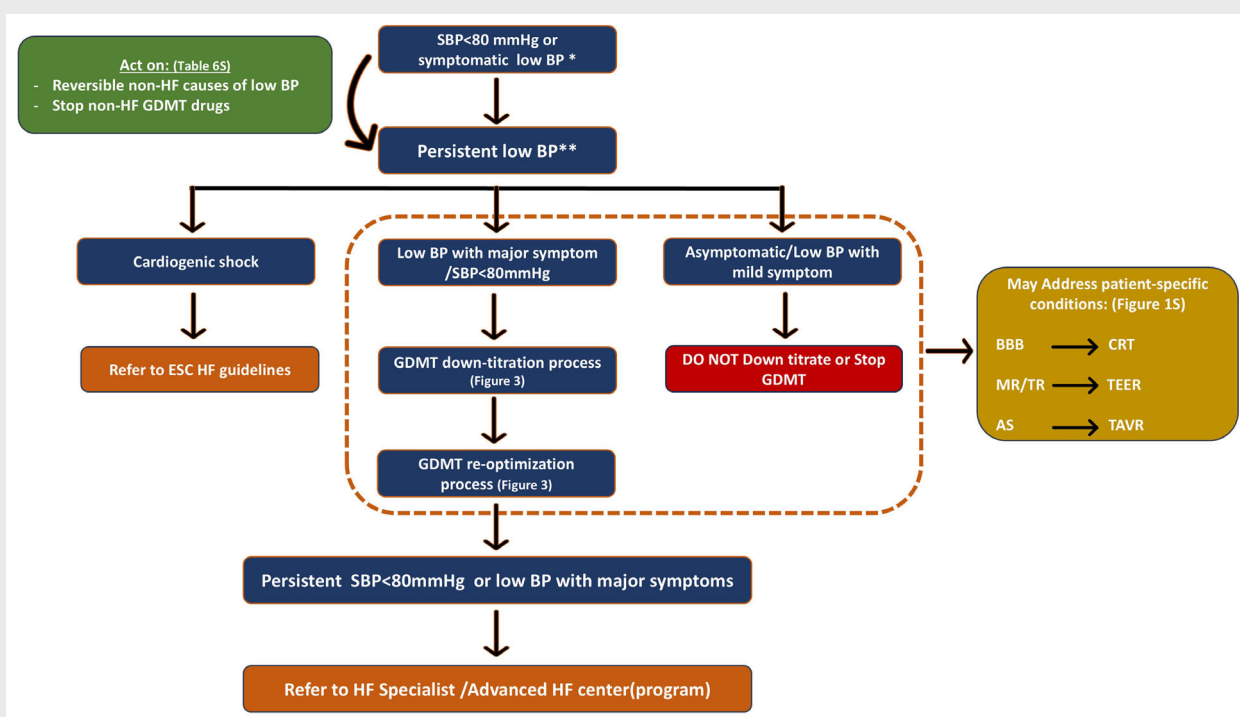
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Despite major advancements in heart failure (HF) management and guideline recommendations over the past two decades, real-world evidence highlights suboptimal implementation of guideline-directed medical therapy (GDMT) for HF with reduced ejection fraction (HFrEF). Low blood pressure (BP) is common in HFrEF patients and represents a major perceived barrier to implementing life-saving treatments in clinical practice, as physicians are often concerned about symptomatic hypotension and its consequences. Although low BP can be seen in those hospitalized with signs of shock, the most common scenario involves non-severe, asymptomatic hypotension in patients receiving foundational therapy for HFrEF, where premature down-titration or discontinuation of GDMT should be avoided. This clinical consensus statement provides a comprehensive overview of low BP in HFrEF, including its definition, risk factors, and effects of HF therapies on BP. We propose management pathways to optimize HFrEF treatment in the context of low BP, ultimately aiming to improve patient outcomes.

*Corresponding author. Division of Cardiology, Balamand University Faculty of Medicine, Balamand, Lebanon. Tel: +96170143142 or +971501720164, Email: skourihadi73@gmail.com

†Contributed equally as co-first authors.

Graphical Abstract



Proposed approach for clinical management and therapeutic optimization in patients with heart failure with reduced ejection fraction and low blood pressure. AS, aortic stenosis; BBB, bundle branch block; BP, blood pressure; CRT, cardiac resynchronization therapy; ESC, European Society of Cardiology; GDMT, guideline-directed medical therapy; HF, heart failure; MR, mitral regurgitation; SBP, systolic blood pressure; TAVR, transcatheter aortic valve replacement; TEER, transcatheter edge-to-edge repair; TR, tricuspid regurgitation. *Low blood pressure with major or mild symptoms. **SBP < 80 mmHg or symptomatic low BP or asymptomatic low BP.

Keywords

Blood pressure • Hypotension • Guideline-directed medical therapy • Treatment implementation • Heart failure with reduced ejection fraction

Preamble

Contemporary management of heart failure (HF) with reduced ejection fraction (HFrEF) has impacted survival, hospitalization rates, and quality of life.^{1–3} Trials have demonstrated early benefit from prompt initiation and large benefits of comprehensive HF therapies over time.^{4–8} Despite several positive landmark trials and guideline recommendations, real-world evidence has shown poor implementation of guideline-directed medical therapy (GDMT), in terms of limited initiation, up-titration, and long-term maintenance.^{5–16} Several barriers hinder the effective implementation of HF guidelines, including factors related to patient/caregiver, physician/provider, and healthcare services.^{17–19} Comorbidities, advanced age, perceived 'polypharmacy', and limited awareness of HF can affect adherence to treatments and contribute to clinical inertia in prescribing and optimizing medications.¹⁷

In this context, low blood pressure (BP) is among the most common perceived barriers to the initiation, implementation, and

escalation of GDMT, as many HF medications can have hypotensive effect.^{16,20} Indeed, a recent international survey conducted by the Heart Failure Association (HFA) of the European Society of Cardiology (ESC) across 91 countries showed that 66% of respondents identified hypotension as a major concern and 31% as a secondary concern.²¹ Often hypotension identified by cardiovascular clinicians should not be a barrier to GDMT initiation or maintenance as it is asymptomatic or associated with minimal symptoms. In the context of no or few symptoms, GDMT for HF should be continued to realize the large clinical benefits. Interestingly, while low BP in patients with HFrEF is linked to a worse prognosis,^{22,23} treatment with GDMT has been shown to weaken this association, highlighting the ongoing beneficial effects of recommended HF therapies even in patients with low BP.²⁴ Major recent landmark HF clinical trials have reported that the relative impact of pharmacological therapy on clinical outcomes was consistent across baseline BP categories considered for inclusion.^{25–30} This apparent contradiction highlights the complexity of HF treatment, underscoring the need

for nuanced and personalized approaches in managing patients with low BP. In this regard, a recent HFA consensus document on GDMT optimization highlighted the importance of patient profiling, which involves tailoring medical therapy based on clinical characteristics such as BP, heart rate (HR), and the presence of comorbidities, to guide the initiation and up-titration of treatment.³¹

This clinical consensus statement from the HFA of the ESC highlights the perception of low BP as a key contributor to clinical inertia and a barrier to implementing HF treatment, summarizes recent evidence on the topic, and proposes strategies for optimizing GDMT in clinical practice among patients with HFrEF and low BP (*Graphical Abstract*).

Prevalence of low blood pressure in the heart failure with reduced ejection fraction population

The prevalence of low BP in the HFrEF population depends on factors such as the severity of HF, the presence of comorbidities, age, gender, and clinical presentation. Additionally, it varies across clinical trials and registries based on the chosen numerical threshold, which is often arbitrary (online supplementary Tables S1–S3), and the specific definition used.^{4,24,32–44} Common definitions include an absolute systolic BP (SBP) value of <90 mmHg or a mean arterial pressure (MAP) of <65 mmHg at rest; the presence of orthostatic hypotension (defined as a sudden drop in SBP of >20 mmHg, or in diastolic BP [DBP] of >10 mmHg upon standing); hypotension (BP <90–100 mmHg) with clinically relevant symptoms (such as dizziness, syncope, headache, visual disturbances, emesis, or fatigue); and hypotension as a side effect or adverse event of GDMT or other medications (online supplementary Table S4). Notably, in symptomatic hypotension, the threshold for low BP is often not clearly defined and may be arbitrarily chosen, as individuals tolerate different levels of low BP differently. However, in clinical trials the inclusion threshold for clinically relevant low BP is almost always fixed (usually close to 90 mmHg) to prevent dropouts and preserve the trial's statistical power. It is also important to note that while 'hypotension' and 'low BP' are semantically synonymous, 'hypotension' is sometimes incorrectly used to describe low BP associated with symptoms. This approach can be misleading, and any instance of low BP accompanied by symptoms should always be clearly stated, as it has significant implications for clinical management.

Real-world data from registries (online supplementary Table S3) shows that around 3–4% of the HFrEF population have a low BP in the outpatient setting (defined as SBP <90 mmHg in the Swedish Heart Failure Registry or <95 mmHg in the CHECK-HF [Chronisch Hartfalen ESC-richtlijn Cardiologische praktijk Kwaliteitsproject Hartfalen] registry).^{11,24} Among 18 677 adult patients monitored between 2000 and 2005 in the Health Improvement Network UK primary care database, the incidence of hypotension (SBP ≤90 mmHg) was 14% over a mean follow-up period of 3.3 years. This corresponds to a rate of 3.17 cases per 100 patient-years, with a notably higher incidence in women aged 18–39 years, who experienced 17.72 cases per 100 patient-years.⁴¹

The prevalence of hypotension increases with the severity of HF. Indeed, in patients with HFrEF hospitalized for acute decompensation, the prevalence of low BP is predictably higher than in the chronic setting (online supplementary Table S3). Across HF registries, the prevalence of low BP ranged from 9% in the ADHERE (Acute Decompensated Heart Failure National Registry),^{42,45} with a SBP cutoff of <90 mmHg, to 22% in the CHAMP-HF (Change the Management of Patients with Heart Failure) registry¹² at SBP <110 mmHg, and 25% in the OPTIMIZE-HF (Organized Program to Initiate Lifesaving Treatment in Hospitalized Patients With Heart Failure) registry,³⁵ with a SBP threshold of <120 mmHg. As expected, in the acute HF population, the incidence of hypotension is consistently linked to the severity of the illness and the treatments being administered.

Prognostic impact of low blood pressure

Low BP is an important prognostic marker in patients with HFrEF, as also demonstrated by its inclusion in several risk models such as the PROMPT-HF (identifying Patients at increased Risk for Poor Outcomes from HFrEF) score, the LIFE-HF (Lifetime prediction and Treatment Benefit in HFrEF) score and the Seattle Heart Failure Model.^{46–48} On the other hand, when patients are on GDMT, the prognostic impact of a low SBP is diminished.²⁴

In the Swedish Heart Failure Registry, the composite risk of cardiovascular death or HF hospitalization rose as SBP decreased, showing a 2.5-fold increase at SBP <80 mmHg and a 1.5-fold increase at SBP <100 mmHg, compared to an SBP of 120 mmHg. Notably, the association between low BP and increased mortality was attenuated in HFrEF patients on optimized GDMT, suggesting that, beyond illness severity, the higher mortality may also relate to fewer patients in this group reaching target dose medications.²⁴ In the QUALIFY (Quality of Adherence to guideline recommendations for Life-saving treatment in heart failure survey) international registry, the association of elevated resting HR and lower SBP identify patients at highest risk for cardiovascular events.⁴⁹

In the acute setting, data from registries indicate that low SBP at admission in patients hospitalized for HF is an independent predictor of increased morbidity and mortality.^{45,50,51} Additionally, low SBP during hospitalization for HF has been identified as an independent predictor of adverse clinical events in patients with HFrEF enrolled in the EVEREST (Efficacy of Vasopressin Antagonism in Heart Failure: Outcome Study with Tolvaptan) trial.⁵²

Effect of heart failure treatments on blood pressure

Several HF treatments can lead to a reduction in BP, with the most significant decreases observed in treatments involving angiotensin receptor–neprilysin inhibitors (ARNi).⁵³ However, it is important to highlight that GDMTs have been proven effective and safe across all baseline SBP levels, and the hypotensive effect of HFrEF medications diminishes as the baseline SBP decreases

(online supplementary Tables S1 and S5). In the SOLOIST-WHF (Effect of Sotagliflozin on Cardiovascular Events in Patients with Type 2 Diabetes Post Worsening Heart Failure) trial, which evaluated the sodium–glucose cotransporter 2 (SGLT2) inhibitor sotagliflozin in 1222 patients experiencing a recent episode of worsening HF, treatment-emergent hypotension was reported in 6% of the treatment arm compared to 4.6% in the placebo arm.⁵⁴ The prevalence of symptomatic hypotensive events was similar in the EMPEROR-Reduced (Empagliflozin Outcome Trial in Patients with Chronic Heart Failure and a Reduced Ejection Fraction) trial (5.7% in the empagliflozin group and 5.5% in the placebo group),⁵⁵ and significantly lower in the DAPA-HF (Dapagliflozin and Prevention of Adverse Outcomes in Heart Failure) trial (0.3% in the dapagliflozin group and 0.5% in the placebo group),⁵⁶ probably due to differences in symptomatic hypotension cut-off and population characteristics (online supplementary Table S1). Notably, in the DAPA-HF trial, patients in the lower baseline BP strata (95–110 mmHg) experienced the smallest average BP decrease with dapagliflozin compared to placebo (–1.50 mmHg), which diminished to less than 1 mmHg after 4 months of treatment.²⁹ The same increasing BP trend was observed in the EMPEROR-Reduced trial among asymptomatic HFrEF patients with lower SBP (i.e. 100–110 mmHg) treated with empagliflozin,²⁵ and in the COPERNICUS (Carvedilol Prospective Randomized Cumulative Survival) trial in patients with SBP 85–95 mmHg treated with carvedilol.⁵⁷ Finally, in the EMPHASIS-HF (Eplerenone in Mild Patients Hospitalization and Survival Study in Heart Failure) trial, patients in the lower strata of baseline SBP (95–105 mmHg) randomized to eplerenone showed an average increase of SBP at 1 and 6 months of 2.8 mmHg compared to placebo,³⁰ likely reflecting an improvement in cardiac function. In the PARADIGM-HF (Prospective Comparison of ARNi with an angiotensin-converting enzyme [ACE] inhibitor to Determine Impact on Global Mortality and Morbidity in Heart Failure) trial, hypotension was a key predictor of non-completion during the run-in phase.⁵⁸ Similarly, hypotension or hypotensive symptoms were the primary reasons for discontinuing low-dose sacubitril/valsartan in the run-in phase of the LIFE (LCZ696 In Hospitalized Advanced Heart Failure) trial.⁵⁹ Among the 8399 patients enrolled in the PARADIGM-HF trial, 1343 (16.0%) experienced asymptomatic hypotension, while 936 (11.1%) experienced symptomatic hypotension at least once after randomization. Importantly, the efficacy and safety of sacubitril/valsartan were maintained, regardless of the presence or absence of hypotension.⁶⁰ This concept was further supported by a post-hoc analysis of PARADIGM-HF, which showed that the benefits of sacubitril/valsartan over enalapril were consistent across all baseline SBP categories (<110, 110 to <120, 120 to <130, 130 to <140, and >140 mmHg),²⁸ reinforcing the idea that low BP should not hinder the optimization of GDMT (online supplementary Tables S1 and S5). Interestingly, HFrEF patients in the lowest baseline SBP strata (95–110 mmHg) experienced a mild BP increase over the entire follow-up period in both study arms, even if this effect was more pronounced in the enalapril group.²⁸

Low BP events during HF treatment were understandably more prevalent in the acute setting than in the chronic setting. In the ASCEND-HF (Acute Study of Clinical Effectiveness of Nesiritide

in Decompensated Heart Failure) trial,⁴³ evaluating the effect of nesiritide in patients hospitalized for acute HF, 22% of patients had an in-hospital episode of hypotension (73.1% asymptomatic), with key factors including randomization to nesiritide, chronic metolazone therapy, and severe illness indicators such as baseline orthopnoea and an S3 gallop. In the PIONEER (Comparison of Sacubitril-Valsartan versus Enalapril on Effect on NT-proBNP in Patients Stabilized from an Acute Heart Failure Episode) trial,⁶¹ several precautions were taken to prevent treatment-induced hypotension, such as maintaining a SBP >100 mmHg for the previous 6 h, ensuring clinical stability (no need for escalating diuretic or inotropic therapy in the last 24 h), and initiating treatment at a low dose for patients with lower SBP. Despite these measures, about 20% of patients in each treatment group discontinued treatment by 8 weeks, predominantly due to adverse events, with symptomatic hypotension occurring in 15% of the treatment group and 12.7% in the placebo group. In the STRONG-HF (Safety, Tolerability and Efficacy of Rapid Optimization, Helped by NT-proBNP Testing, of Heart Failure Therapies) study,⁴ which aimed to evaluate rapid therapeutic implementation in selected patients with recent acute HF hospitalization but haemodynamically stable and with elevated natriuretic peptides at the time of randomization, hypotension occurred in 5% of the high-intensity implementation group compared to 1% in usual care, despite strict follow-up visits. Additionally, orthostatic hypotension was reported in 0.6% of the high-intensity implementation group versus none in usual care, while dizziness occurred in 0.7% and 0.4%, respectively.

In the VICTORIA (Vericiguat Global Study in Subjects With Heart Failure With Reduced Ejection Fraction) trial, which included patients with HFrEF, a recent worsening HF and a baseline SBP >100 mmHg, those with an initial SBP ≥110 mmHg experienced a more pronounced initial decline in SBP with vericiguat compared to placebo over the first 16 weeks (especially patients older than 75) before returning to baseline levels. Conversely, patients with a baseline SBP <110 mmHg showed a trend of increasing SBP over time in both study arms.²⁷ In the GALACTIC-HF (Global Approach to Lowering Adverse Cardiac Outcomes through Improving Contractility in Heart Failure) trial, which included inpatient and outpatient with HFrEF and a SBP ≥85 mmHg, omecamtiv mecarbil significantly reduced the risk of HF outcomes in those with lower baseline SBP (≤100 mmHg) compared with placebo, without affecting BP values.^{62,63}

Low blood pressure in the clinical setting

Low BP in patients with HF may occur with or without symptoms and can happen in both inpatient and outpatient settings, though it is more commonly observed during hospitalizations for HF, as previously mentioned. Hypotension can occur in various clinical contexts, including among patients new to HF treatment, during the initiation of therapy, and among those receiving optimal medical management.⁴¹ In each scenario, it is essential to pinpoint the specific circumstances and timing of the low BP episodes, especially in relation to HF treatment, to identify and address all factors

Table 1 Cardiovascular and non-cardiovascular potential causes of low blood pressure

Cardiovascular-related	Non-cardiovascular-related
- Advanced HF (low output state-NYHA IIIb–IV; hyponatraemia) - Decrease blood volume (recent bleed, trauma ...) - Valve disease (significant MR, TR, and AS) - Bradycardia (AV blocks ...) - Dehydration (over diuresis; high temperature, diarrhoea) - An increase in diuretic dose in the previous week - Prolonged bed rest (OH) - Pregnancy 1st 24 weeks - BP-lowering medications: CCB; alpha-blockers, centrally acting; diuretics (thiazides ...); ACEis; ARBs; BB - Angina medications: BB; CCB; nitrates - HF non class I: hydralazine/nitrates - Neurally mediated syncope - Altered vasoreactivity related to comorbidities (diabetes) Acute—rapidly progressive - Peripartum cardiomyopathy - Myocarditis - Intraabdominal hypertension (ascites) - Persistent signs of congestion requiring high doses of diuretics or vasodilators	- Elderly (higher age group) - CKD - Endocrine problems: hypothyroidism; adrenal insufficiency (Addison's disease); hypoglycaemia; diabetes - Infection (when severe): UTI, septic shock - Allergic reaction (anaphylaxis) - Nutrient deficiency: vitamin B12 and folate causing anaemia - Iron deficiency - Liver disease: hepatic cirrhosis - COPD - Dementia - Parkinson's - Depression - Medications (see online supplementary Table S6) - Antidepressants - Antipsychotics (2nd generation) - Dopamine antagonist - PDE5 inhibitors (erectile dysfunction) - Alfa-blockers (BPH) - Complementary and alternative medicine - Narcotics and alcohol - Intraocular beta-blockers (glaucoma)

ACEi, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor antagonist; AS, aortic stenosis; AV, atrio-ventricular; BB, beta-blocker; BPH, benign prostatic hypertrophy; CCB, calcium channel blocker; CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease; MR, mitral regurgitation; NYHA, New York Heart Association; OH, orthostatic hypotension; PDE5, phosphodiesterase 5; TR, tricuspid regurgitation; UTI, urinary tract infection.

contributing to the hypotension. Indeed, low BP can arise from reversible factors such as over diuresis or concomitant other medications that are amenable to intervention or from irreversible factors like the stage of HF and the patient's age. Obviously, the identification of reversible factors is crucial, since it can allow its treatment and the subsequent optimization of GDMT. These contributors may stem from cardiovascular or non-cardiovascular issues, including medication effects and other medical conditions. Identifying the underlying causes of low BP is critical, especially in acute settings where prompt intervention is often required. Factors contributing to low BP in HF can be categorized under two distinct headings (Table 1):

1. Cardiovascular-related: encompassing factors associated with disease severity and cardiovascular therapy, including advanced New York Heart Association (NYHA) class, moderate to severe valvular disease, cardiovascular medications (e.g. antihypertensives, nitrates, diuretics), and conditions like myocarditis and acute ischaemia that exacerbate HF decompensation by reducing stroke volume and/or increasing filling pressures. Notably, in the acute setting, the clinical significance of hypotension often points to reduced cardiac output and stroke volume, despite heightened sympathetic activation, which would typically increase peripheral resistance and cardiac contractility.

2. Non cardiovascular-related: including some comorbidities, such as infections and Parkinson's disease,^{64,65} or non-cardiovascular medical treatments, such as alpha-blockers in prostatic hyperplasia, phosphodiesterase-5 inhibitors for erectile dysfunction, serotonin and norepinephrine reuptake inhibitor and tricyclic/tetracyclic antidepressants in depression, levodopa for Parkinson's disease (online supplementary Table S6).⁶⁶

Notably, these factors can overlap and synergistically contribute to low BP in various pathological states, sometimes leading to misinterpretations as indicators of worsening HF. For instance, a brief assessment may overlook the complexities in a diabetic patient with HF, where low BP could be mistakenly attributed to worsening HF rather than autonomic dysfunction caused by the long-standing course of the disease.⁶⁷

Effect of devices, interventional procedures, and non-class I drugs on heart failure treatment optimization

Heart failure-related devices and interventional procedures (online supplementary Figure S1) may be considered earlier in certain

cases where optimizing medical treatment for HFrEF is challenging due to low BP. These interventions have the potential to improve outcomes, alleviate symptoms, enhance quality of life (QoL), and ultimately increase BP, thus facilitating the prescription of evidence-based therapy. Although current data remain insufficient to confirm whether optimizing GDMT after such interventions leads to better outcomes, it is worth to always reconsider the possibility for optimizing GDMT following these procedures.

Devices and interventional procedures

Cardiac resynchronization therapy

Cardiac resynchronization therapy (CRT) has proven highly effective in improving QoL and reducing HF hospitalizations and mortality in selected patients with HFrEF and left ventricular electrical dyssynchrony.^{1,68–76} Implantation of CRT in this context typically results in an average BP increase of 5%,⁷⁷ primarily due to enhanced myocardial synchrony and improved left ventricular ejection fraction from reverse remodelling. Additionally, this therapy may enable the up-titration of beta-blockers in patients experiencing symptomatic bradycardia.⁷⁸

Transcatheter valve interventions and other procedures

Valvular heart disease, such as aortic stenosis (AS) or mitral/tricuspid regurgitation, in the context of HFrEF may induce or worsen hypotension mainly reducing the forward stroke volume and cardiac output.^{79,80}

Relief of AS has been shown to significantly increase BP after the procedure. Indeed, in the PARTNER I (Placement of Aortic Transcatheter Valve) trial, which focused on patients with severe, symptomatic AS at high or prohibitive surgical risk undergoing transcatheter aortic valve replacement (TAVR), 55% of participants experienced an increase in SBP 30 days post-procedure. Over 6 months post-TAVR, there was a moderate rise in SBP of <10 mmHg and a slight decrease in DBP of <5 mmHg.⁸¹ Another study involving 105 patients with severe AS reported an immediate post-TAVR increase in SBP (average 15 ± 31 mmHg), sustained over 5 days in 51% of the patients.⁸² While there are no specific data for HFrEF settings, it is biologically plausible that similar benefits in BP improvement could be observed in these patients.

Similarly, transcatheter edge-to-edge repair (TEER) of the atrioventricular valves can reduce backflow in the pulmonary or systemic circulation, thereby alleviating congestion and enhancing forward cardiac flow and perfusion. This improvement can lead to beneficial effects on BP and renal function,^{83,84} potentially facilitating the optimization of GDMT. Notably, a recent analysis from the EuroSMR (European Registry of Transcatheter Repair for Secondary Mitral Regurgitation) registry, involving a large, unselected real-world cohort of patients with severe mitral regurgitation and HFrEF, showed that the most significant independent predictor of medical therapy intensification post-mitral TEER was the optimal reduction of mitral regurgitation grade.⁸⁵

Atrial fibrillation (AF) can adversely affect HF. The absence of the atrial kick, poor HR control, impaired adaptation of HR

to metabolic demands, and irregularity in heart rhythm may contribute to decompensation and lower BP. Previous observations have shown that an irregular sequence of RR intervals can decrease cardiac output by approximately 0.8 L/min, independent of HR.⁸⁶ In patients with HFrEF, when AF is suspected to significantly contribute to low BP, restoring sinus rhythm may be a reasonable approach.

Other drugs

Ivabradine

The sinoatrial node I_f -channel inhibitor ivabradine effectively lowers elevated HR in patients with sinus rhythm without adversely affecting BP,²⁶ which may instead show a slight increase.⁸⁷ Ivabradine can be particularly beneficial in patients with HFrEF where low BP limits the optimization or escalation of beta-blockers, potentially enabling more effective beta-blocker management over time, both in the chronic and acute HF scenarios.^{88–90} Notably, a secondary analysis of SHIFT (Systolic Heart Failure Treatment with the I_f Inhibitor Ivabradine Trial) demonstrated that reducing resting HR with ivabradine is effective in improving clinical outcomes for HFrEF patients with low SBP.⁹¹

Digoxin

Digoxin, a cardiac glycoside with inotropic properties and neurohormonal modulating effects by resetting baroreceptor sensitivity with subsequent parasympathomimetic and antiadrenergic effects, resulted in increased cardiac output, afterload reduction and lower pulmonary capillary wedge pressure in HFrEF.⁹² Current guidelines recommend its use in patients with HFrEF as a second-line treatment for AF rate control in combination with beta-blockers.¹ Given that digoxin does not decrease (or even increase) BP,^{93,94} it may be considered for patients with HFrEF, AF and elevated HR, where optimization of GDMT is limited by low BP, to help enhance the optimization of other recommended treatment. A randomized trial is currently underway to evaluate the impact of adding low-dose digoxin to standard care for reducing mortality and HF events in patients with HFrEF or HF with mildly reduced ejection fraction, whether in sinus rhythm or AF (DECISION [Digoxin Evaluation in Chronic heart failure: Investigational Study In Outpatients in the Netherlands]; NCT03783429).⁹⁵ Additionally, another ongoing randomized controlled trial is investigating the efficacy and safety of digitoxin as an adjunct to contemporary HF treatment in patients with advanced HFrEF (DIGIT-HF [DIGitoxin to Improve ouTcomes in patients with advanced chronic Heart Failure]; EudraCT Number: 2013-005326-38).⁹⁶

Pragmatic approach to low blood pressure

Given the limited value of BP as a direct indicator of cardiac function and organ perfusion, clinical decision-making should not be overly influenced by low BP readings unless they are critically low (e.g. SBP < 80 mmHg), indicating significant haemodynamic

Table 2 How to determine if low blood pressure is clinically significant

1. Office BP measurement
 - Obtain BP in the supine/sitting and standing position (for 3 min) to determine if the patient has orthostatic hypotension (drop of 20 mmHg in systolic BP and/or 10 mmHg in diastolic BP)
 - If symptoms during confirmed orthostatic hypotension => Symptomatic hypotension
2. ABPM measurements
 - Consider ABPM if orthostatic hypotension is not identified with office BP measurement
 - Correlate symptoms with low BP episode on ABPM, if correlated => Symptomatic hypotension
3. Reassessment
 - If initial office BP (+/- ABPM) did not identify symptomatic hypotension, consider repeated evaluation during follow-up
4. In all cases (even if no symptoms related to hypotension), address factors triggering hypotension unrelated to HFrEF:
 - a. transient medical conditions (e.g. diarrhoea or fever)
 - b. the use of cardiovascular treatments not recommended for HFrEF

ABPM, ambulatory blood pressure monitoring; BP, blood pressure; HFrEF, heart failure with reduced ejection fraction.

compromise. Instead, patient evaluation should focus on assessing symptoms and/or organ perfusion (online supplementary Figure S2), as low BP does not always correlate with impaired perfusion.⁹⁷

A SBP <80 mmHg or hypotension causing major symptoms in patients with chronic HFrEF, irrespective of the setting, is a critical threshold that warrants careful attention and may necessitate a re-evaluation of medical treatment, including a potential adjustment of GDMT. Hypotension associated with minor symptoms is not a reason to withhold or reduce HF GDMT.

We suggest comprehensive patient assessment instead of reliance on BP metrics alone, ensuring that the management of low BP is tailored to the individual's clinical presentation and the specific demands of acute or chronic settings.

Low blood pressure in patients with chronic heart failure

After confirming adequate organ perfusion (i.e. the absence of cardiogenic shock), the initial step should be to verify BP readings and, if major symptoms are present, assess whether they correspond to the low BP values (Table 2). Persistent poor organ perfusion (possibly accompanied by severe worsening renal function) should trigger a comprehensive evaluation and might necessitate hospitalization or rapid referral to an advanced HF programme (Figure 1).

Initial steps in the management of low blood pressure

Due to the lack of specificity of symptoms indicative of hypotension (i.e. dizziness and fatigue), it is essential to establish a connection

between their occurrence (particularly when standing or in an upright position) and low BP (Table 2), as previously proposed.⁹⁸ This involves measuring BP both while supine and standing.⁹⁹ A drop of 20 mmHg in SBP and/or 10 mmHg in DBP within the first 3 min of standing suggests a BP-related cause for the symptoms. If orthostatic hypotension is not confirmed, ambulatory BP monitoring (ABPM) can be used to identify hypotensive episodes that correlate with reported symptoms, providing also important prognostic information.¹⁰⁰ In cases where SBP levels do not align with symptoms of postural or orthostatic hypotension, it is advisable to re-assess the patient in a short timeframe utilizing ABPM and standing BP assessments as described. A further step would involve identifying factors causing hypotension unrelated to HFrEF, such as transient medical conditions that lead to dehydration (e.g. diarrhoea, fever, or overtreatment with diuretics) or the use of cardiovascular treatments not recommended for HFrEF, such as calcium channel blockers, centrally acting antihypertensive drugs, or alpha-blockers, which should be reduced or discontinued. The following steps differ depending on whether the patient is naïve to treatment or already on GDMT.

Low blood pressure in ambulatory patients with heart failure naïve to therapy

Patients with HFrEF may exhibit low BP at their initial presentation, even before the start of GDMT. Notably, persistent naïve low SBP or intolerance to GDMT may underlie a more advanced HF stage.^{1,101} However, low BP alone neither defines nor is included in the major criteria for diagnosing advanced HF (stage D, according to the American College of Cardiology/American Heart Association criteria).¹⁰¹

If a patient's BP normalizes after initial interventions (modulation or discontinuation of non-HF hypotensive treatments), GDMT should be promptly commenced according to HF guidelines. For persistent low BP in asymptomatic or mildly symptomatic patients with HFrEF and adequate perfusion, a low-dose sequence of HF medications may be started. Since SGLT2 inhibitors and mineralocorticoid receptor antagonists (MRAs) have the least effect on BP, but a rapid beneficial effect,^{102,103} these should be initiated as first drug class. Subsequently, consider starting a low-dose beta-blocker (if HR is >70 bpm) or a low (i.e. 50 mg twice daily) or very low (i.e. 25 mg twice daily) dose of sacubitril/valsartan,¹⁰⁴ then gradually up-titrate if possible. In case of poor tolerance to sacubitril/valsartan, a low-dose ACE inhibitor (or angiotensin receptor blocker [ARB], if ACE inhibitors contraindicated) should be considered. Selective β_1 receptor blockers may also be preferred due to lesser BP-lowering effect than non-selective beta-blockers with α , β_1 and β_2 -blocker or vasodilatory properties. If beta-blockers are not well tolerated haemodynamically, ivabradine may be a viable alternative, either alone or with low-dose beta-blockers, as it helps facilitate their titration.^{88,105,106} Close monitoring is essential, applying a strategy to initiate or up-titrate one drug at a time using small increments until the highest tolerated or target dose of each is achieved (Figure 1). Diuretics will need to be adjusted according to volume status and may result in lower BP with overdiuresis.

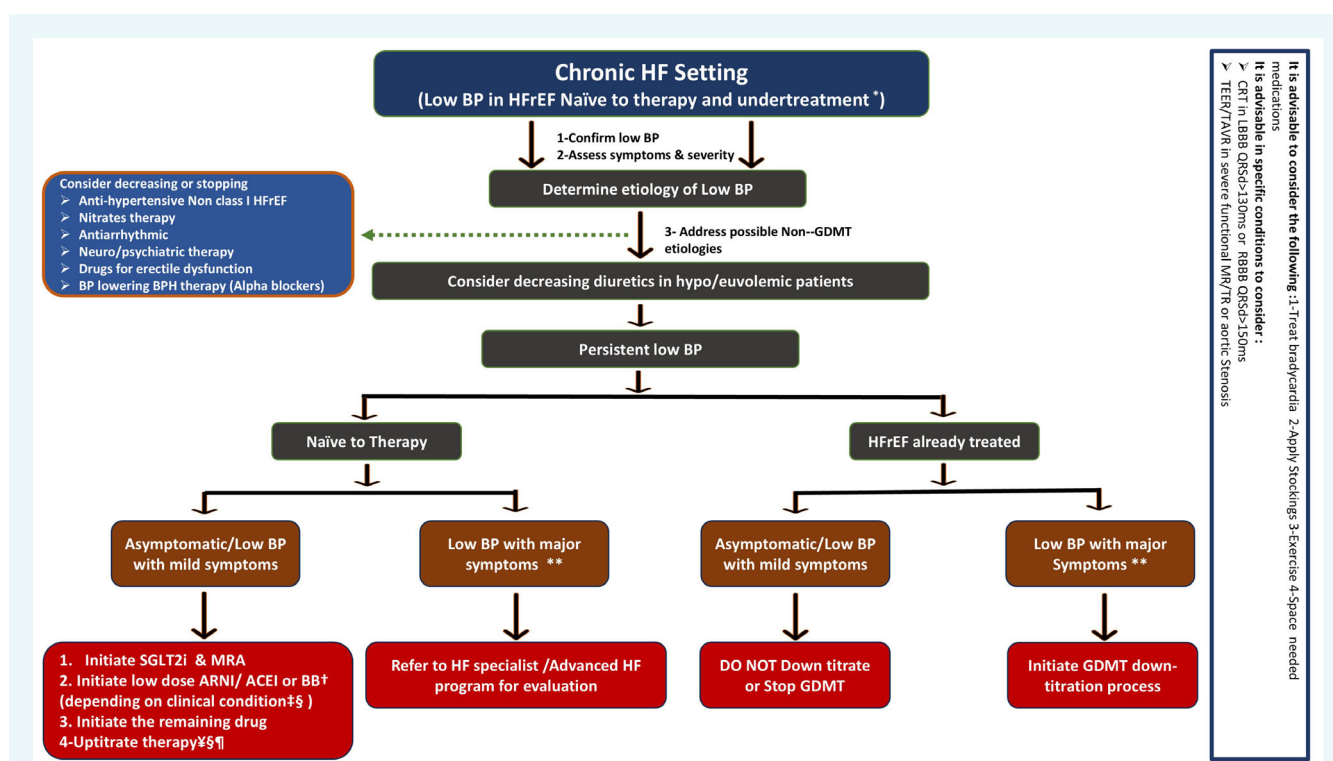


Figure 1 Proposed algorithm for treatment optimization in patients with heart failure with reduced ejection fraction and low blood pressure, both therapy-naïve and already on treatment. ACEI, angiotensin-converting enzyme inhibitor; ARNI, angiotensin receptor–neprilysin inhibitor; BB, beta-blocker; BP, blood pressure; BPH, benign prostatic hypertrophy; CRT, cardiac resynchronization therapy; GDMT, guideline-directed medical therapy; HF, heart failure; HFrEF, heart failure with reduced ejection fraction; LBBB, left bundle branch block; MR, mitral regurgitation; MRA, mineralocorticoid receptor antagonist; QRSd, QRS duration; RBBB, right bundle branch block; SGLT2i, sodium–glucose cotransporter 2 inhibitor; TAVR, transcatheter aortic valve replacement; TEER, transcatheter edge-to-edge repair; TR, tricuspid regurgitation. *If patient stable on optimal GDMT it is unlikely related to HFrEF therapy and look for other aetiologies; while if recently GDMT was initiated or up-titrated then most likely related to one of the GDMT drugs. **SBP <80 mmHg; significant orthostatic hypotension, fatigue, tiredness, dizziness, etc. †Start lowest dose and up-titrate slowly with small increments whether for renin–angiotensin–aldosterone system or BB. ‡Since blood pressure is low, may tolerate low dose sacubitril/valsartan if not initiate low-dose ACEI/and try up-titrate (with time may be able to shift to sacubitril/valsartan). §Start SGLT2i and MRA in the first step as they do not lower BP, then consider either low-dose BB if heart rate >70 bpm or ARNI/ACEI/ARB at low dose and then up-titrate weekly with low increments till reach target dose or highest tolerated dose. Sometimes BP improves and medications can be up-titrated more easily (increase in cardiac output). If the patient achieves an adequate dose of ACEI or ARB (tolerating up-titration), then can be shifted to low-dose sacubitril/valsartan. ¶If cannot tolerate even low dose of BB and in sinus rhythm, ivabradine may be used and later BB can be initiated or up-titrated. In atrial fibrillation and uncontrolled heart rate, digoxin maybe used. ††Up-titrate small increments every 1–2 weeks one drug at a time and close observation and follow-up.

Low blood pressure in ambulatory patients with heart failure under treatment

In the ambulatory setting, low BP during GDMT is quite frequent in patients with HFrEF and does not always indicate poor tolerance to HF drugs.^{24,32,36} Symptomatic hypotension in chronic HFrEF, typically characterized by mild dizziness upon standing, can usually be managed through patient education and counselling, without necessitating a reduction in HF pharmacotherapy.^{36,39} Patients often feel reassured and remain compliant to their treatment regimen when they understand that transient dizziness is a side effect of life-prolonging HF drugs which also reduce hospitalizations and enhance QoL.^{1,3,107}

As a rule of thumb, patients who are clinically stable on optimal medical therapy but with low BP, whether symptomatic or

asymptomatic, are unlikely to have their condition directly caused by GDMTs. This common scenario should promptly lead to an evaluation for other potential causes, both cardiovascular (e.g. valvular diseases or myocardial ischaemia) and non-cardiovascular (e.g. initiation of alpha-blockers for benign prostatic hyperplasia) to avoid unnecessary interruptions, reductions, or discontinuations of foundational HFrEF therapies, as outlined above. The first step in this setting should be to assess the congestion status to assess the feasibility of a possible diuretic dose reduction. Clinical, biological, or ultrasound (lung and/or cardiac) signs of congestion should be explored to attain this goal.^{108,109} In the absence of congestive signs, diuretics could be cautiously decreased. Although only a few trials have focused on reducing diuretics,^{110,111} a recent randomized controlled trial has shown that this approach is feasible in stable

patients with HFrEF.¹¹² Serial monitoring of stable natriuretic peptide levels can be useful during diuretic titration to ensure that congestion does not worsen significantly.¹¹³ If adjusting diuretics fails or is not feasible, and clinically relevant symptomatic hypotension persists, it may be necessary to reconsider the dosages or types of other HFrEF medications. This is particularly important for patients undergoing therapy optimization or up-titration, where low BP is most likely due to the initiation or increase of HF medications (Figure 1).

It is crucial for general physicians and cardiologists to recognize that certain HF medications, such as SGLT2 inhibitors and MRAs,^{29,30} rarely cause low BP, while others, like sacubitril/valsartan, are more likely to lead to symptomatic hypotension.⁵³ This understanding can guide physicians in effectively prioritizing and sequencing treatments for patients with chronic HFrEF and low BP.³¹

Low blood pressure in patients with acute heart failure

In the acute HF setting, BP values should be interpreted within the context of the patient's clinical presentation and trajectory. In a recent acute HF study, patients with low SBP and no signs of hypoperfusion had a similar prognosis as patients with normal SBP.¹¹⁴ Indeed, clinical profiling based on congestion and perfusion status is crucial as BP alone is insufficient to guide therapy and predict prognosis.

Most patients in the acute setting are congested and well perfused ('wet and warm' phenotype of Forrester classification),¹¹⁵ thus showing no contraindication to either continue or initiate GDMT after reaching clinical stability.^{1,116–119} In cases of acute HF with low BP, it is crucial to exclude cardiogenic shock (defined by SBP <90 mmHg coupled with signs of organ hypoperfusion), as the treatment approach will follow the respective guidelines.^{1,120} Notably, it should be borne in mind that during the initial phases of cardiogenic shock, hypoperfusion may occur without hypotension, as BP may be maintained through compensatory vasoconstriction at the expense of tissue perfusion and oxygenation.^{50,121}

During the initial 48 h of hospital admission, the primary goals are to stabilize patient haemodynamically, treat volume overload and congestion, and ensure adequate tissue oxygenation. At this stage, the continuation or initiation of GDMT should be evaluated according to clinical phenotype. Typically, in the 'dry and cold' and 'wet and cold' acute HF phenotypes, HF therapy may be withheld or reduced, especially beta-blockers and/or renin–angiotensin system inhibitors. After the first 48 h, HF treatment can be initiated, reinitiated, or up-titrated based on the evolving clinical scenario and the specific patient characteristics (i.e. optimal or suboptimal GDMT, de novo acute HF, etc.). Typically, it is best to start or restart medications that have the least impact on BP (i.e. SGLT2 inhibitors or low-dose MRAs), then gradually introduce other drugs through conservative up-titration. This process of increasing dosage might extend over 4–6 weeks, requiring close monitoring and small incremental adjustments, with each drug added sequentially. All efforts should be made to initiate any missing foundational therapy, unless contraindicated (Figure 2).

This early optimization-focused protocol is supported by the findings of the STRONG-HF trial, which demonstrated that an intensive treatment strategy, involving the up-titration of GDMT to 100% of recommended doses within 2 weeks of discharge and close follow-up after acute HF admission, significantly reduced symptoms, improved QoL, and decreased all-cause mortality or HF readmissions at 180 days compared to a usual care strategy.⁴ Although the presence of non-cardiovascular comorbidities or advanced age did not restrict the rapid, high-intensity up-titration of HF therapies or diminish the benefits on the primary endpoint,¹²² the evidence of low baseline BP increased the likelihood of meeting safety indicators used to guide up-titration (e.g. estimated glomerular filtration rate [eGFR] <30 ml/min/1.73 m², serum potassium >5.0 mmol/L, SBP <95 mmHg, HR <55 bpm, and N-terminal pro-B-type natriuretic peptide concentration >10% higher than pre-discharge values).¹²³ This resulted in lower administration of GDMT doses, a reduced average percentage of optimal GDMT doses, and less improvement in QoL, although the benefits of an intensive treatment strategy in reducing morbidity and mortality persisted regardless of baseline SBP.¹²³ Consequently, patients recovering from acute HF hospitalization who present with lower BP may require a more gradual up-titration protocol and closer monitoring after discharge to ensure optimal HF treatment.

Specific clinical conditions

In patients with HFrEF and low BP, it is crucial to recognize that when certain clinical conditions coexist, low BP only, symptomatic hypotension or evidence of hypoperfusion can be particularly detrimental and may adversely affect outcomes. These include significant peripheral vascular disease especially with non-healing ulcers and wounds, significant bilateral untreated carotid stenosis, recent cerebrovascular accident (transient ischaemic attack or stroke), bowel ischaemia or abdominal angina, end-stage renal disease on dialysis, and autonomic dysfunction condition. Effective management in these scenarios requires a multidisciplinary team approach to improve patient outcomes.

How to sequence drug discontinuation in different scenarios

When all strategies to manage low BP in patients with HFrEF have been addressed (stopping, reducing, or changing 2021 HF guidelines' not indicated treatments,¹ comorbidity therapies, and other non-proven beneficial drugs) and major symptoms or severely low BP (<80 mmHg) persist, reducing or discontinuing one or more of GDMT drugs may be necessary. It is important to recognize that the poor outcomes associated with the side effects of HF medications may stem from discontinuing the therapy rather than the side effects themselves. In a real-world outpatient HF setting, Rossignol¹²⁴ analysed data from the ESC HFA EURObservational Research Programme (EORP) Heart Failure Long-Term Registry to explore the relationship between hyperkalaemia, the discontinuation of renin–angiotensin–aldosterone system inhibitors (RAASi), and associated outcomes. While hyperkalaemia poses a significant challenge to the initiation and continuation of RAASi therapy, it

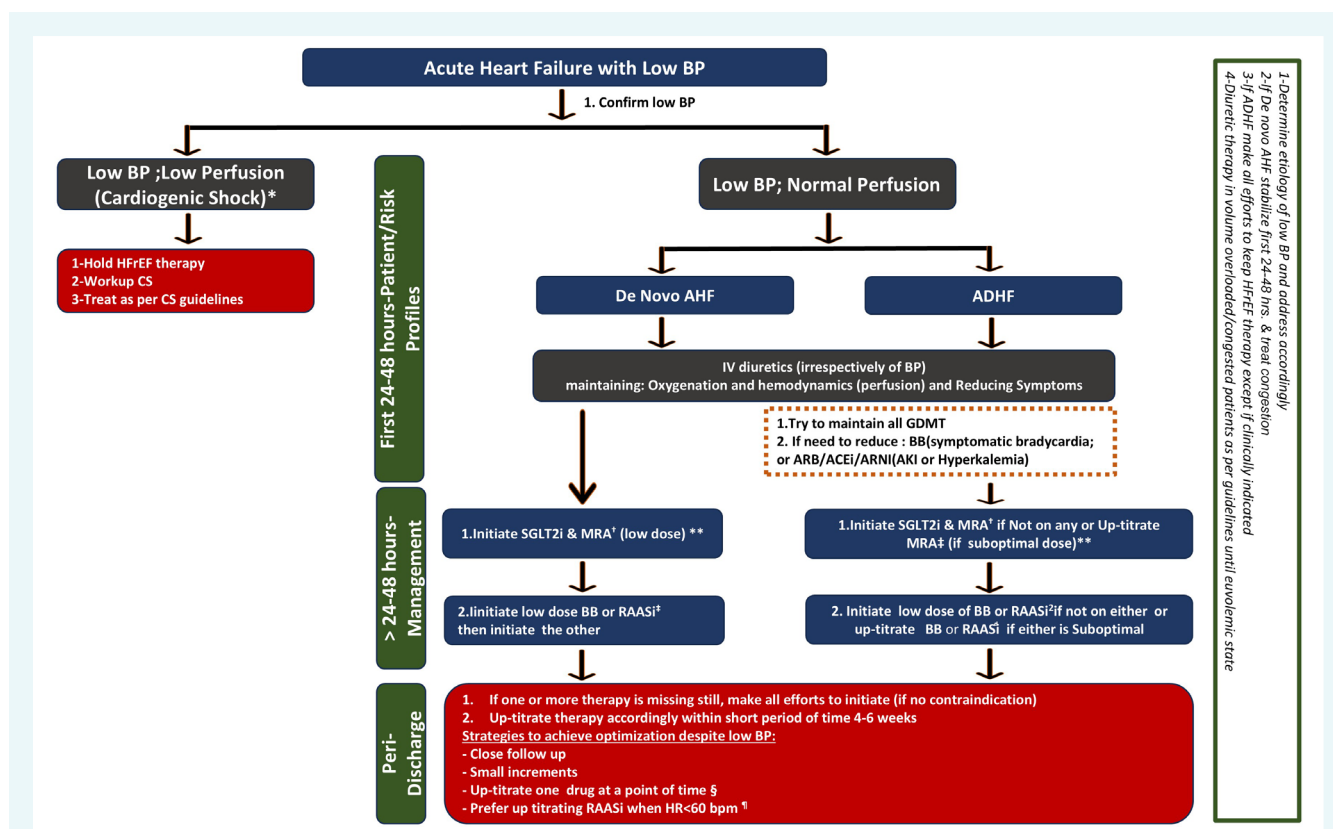


Figure 2 Proposed algorithm for treatment optimization in patients with heart failure with reduced ejection fraction and low blood pressure in the acute heart failure setting. ACEi, angiotensin-converting enzyme inhibitor; ADHF, acute decompensated heart failure; AHF, acute heart failure; AKI, acute kidney injury; ARB, angiotensin receptor blocker; ARNI, angiotensin receptor–neprilysin inhibitor; BB, beta-blocker; BP, blood pressure; CS, cardiogenic shock; GDMT, guideline-directed medical therapy; HFrEF, heart failure with reduced ejection fraction; HR, heart rate; IV, intravenous; MRA, mineralocorticoid receptor antagonist; RAASi, renin–angiotensin–aldosterone system inhibitor; SGLT2i, sodium–glucose cotransporter 2 inhibitor. *Clinical profile: wet/cold or dry/cold. **Estimated glomerular filtration rate >30 ml/min/1.73 m² for MRA or >20 ml/min/1.73 m² for SGLT2i. †If potassium <5.0 . ‡Preferably initiate low-dose ARNI if not tolerated shift to ACEi/ARB. §If rate uncontrolled not tolerating BB and in sinus rhythm may consider ivabradine and in atrial fibrillation consider digoxin. ¶Initiate very low-dose ARNI (25 mg twice daily) and up-titrate, if not tolerated shift to low-dose ACEi/ARB.

may also serve as a risk marker contributing to therapy discontinuation and suboptimal use, ultimately resulting in higher mortality rates. In an analysis of TOPCAT-Americas, patients randomized to spironolactone were more likely to receive lower doses and to discontinue the study drug with a rate of up to 30% during the first year among those with hyperkalaemia and low glomerular filtration rate. Notably, spironolactone discontinuation was associated with a two to fourfold higher risk of subsequent adverse events.¹²⁵ In an analysis of the Observational study from the Stockholm CREAtinine Measurements (SCREAM) project, stopping an MRA after an episode of hyperkalaemia was associated with reduced risk for recurrent hyperkalaemia, but higher risk of death or cardiovascular events.¹²⁶ In situations requiring sequenced drug discontinuation, a thorough understanding of the patient's clinical characteristics or phenotype is crucial for informed decision-making (Figure 3). During this process, all efforts should be concentrated on continuing treatment with SGLT2 inhibitors and potentially MRAs (if not contraindicated), as these drugs have the least impact on BP. Concurrently, to alleviate symptoms of low BP and maintain a

higher percentage of GDMTs, non-pharmacological interventions should be implemented. These include spacing out medications to reduce the synergistic hypotensive effect of HF medications, engaging in exercise and physical training which are proven to improve both orthostatic hypotension and low BP, and using compression leg stockings to help minimize orthostatic drops in BP due to orthostatic changes.^{127–131}

Since real-world evidence suggests that discontinuing or failing to restart HF therapy can significantly increase mortality,^{7,124–126} it is essential to reintroduce treatment as soon as possible. Some practical tips can help facilitate the initiation or reinitiation of GDMT (Figure 3):

- Always reintroduce medications at the lowest possible dose and carefully up-titrate with close monitoring, avoiding simultaneous reinitiation of two drugs.
- Initiate treatment with a low or very low dose of ARNi. If not tolerated, switch to a low dose of ACE inhibitors or ARBs, which have a less pronounced impact on BP.

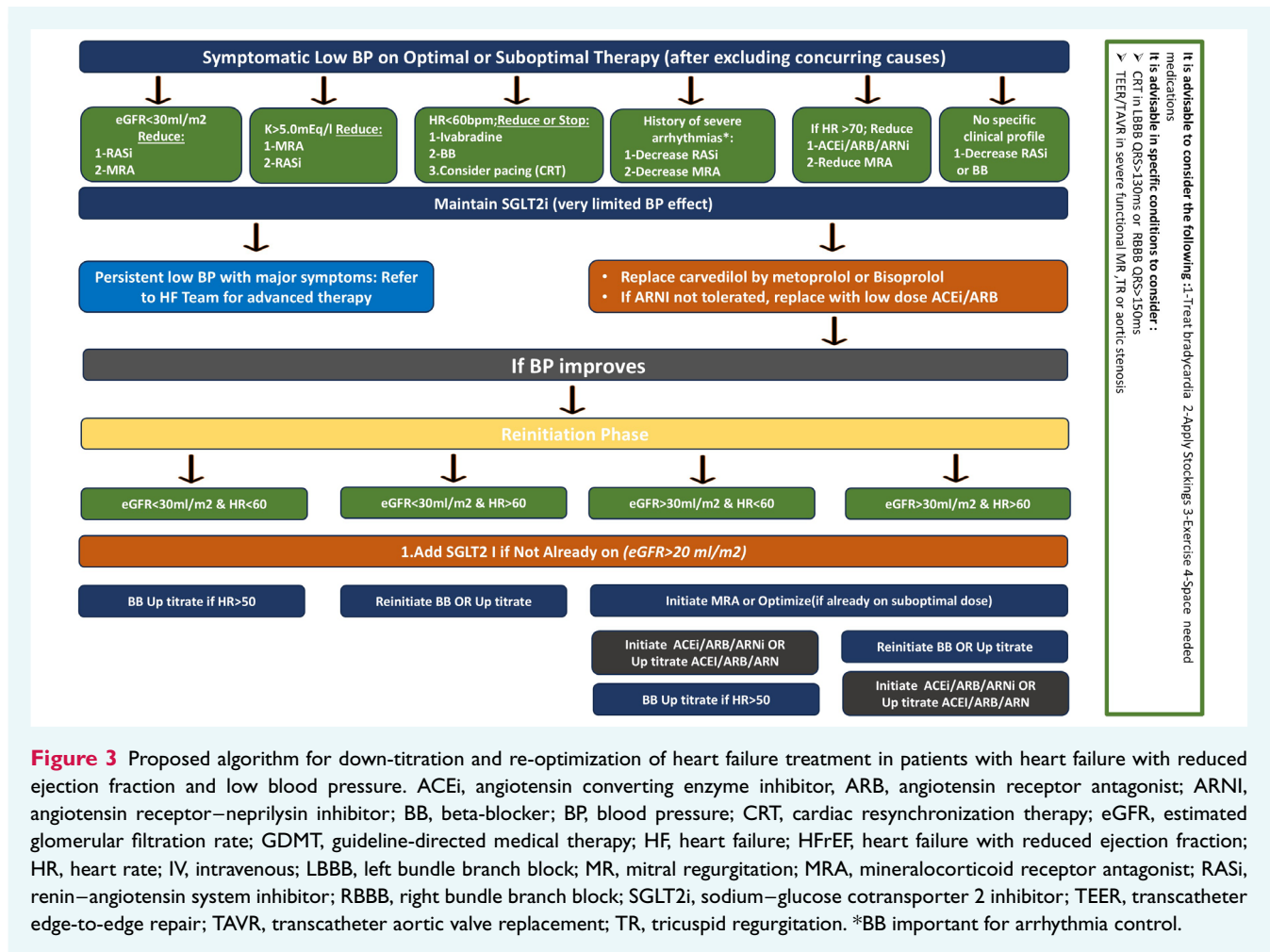


Figure 3 Proposed algorithm for down-titration and re-optimization of heart failure treatment in patients with heart failure with reduced ejection fraction and low blood pressure. ACEi, angiotensin converting enzyme inhibitor; ARB, angiotensin receptor antagonist; ARNI, angiotensin receptor–neprilysin inhibitor; BB, beta-blocker; BP, blood pressure; CRT, cardiac resynchronization therapy; eGFR, estimated glomerular filtration rate; GDMT, guideline-directed medical therapy; HF, heart failure; HFrEF, heart failure with reduced ejection fraction; HR, heart rate; IV, intravenous; LBBB, left bundle branch block; MR, mitral regurgitation; MRA, mineralocorticoid receptor antagonist; RASi, renin–angiotensin system inhibitor; RBBB, right bundle branch block; SGLT2i, sodium–glucose cotransporter 2 inhibitor; TEER, transcatheter edge-to-edge repair; TAVR, transcatheter aortic valve replacement; TR, tricuspid regurgitation. *BB important for arrhythmia control.

- Consider switching from carvedilol or nebivolol to a selective beta-blocker without vasodilatory properties, such as metoprolol or bisoprolol.
- If the target HR (<70 bpm) is not achieved and increasing the dose of a beta-blocker is not feasible, consider adding ivabradine for patients in sinus rhythm or digoxin for those with AF.
- For all patients with HFrEF, low BP and an eGFR > 20 ml/min/1.73 m², initiate treatment with an SGLT2 inhibitor. If the patient is already on an SGLT2 inhibitor, consider up-titrating an MRA if eGFR is >25–30 ml/min/1.73 m², or prioritize beta-blockers if eGFR is <25–30 ml/min/1.73 m².
- If therapy reintroduction results in persistent symptomatic low BP and/or hypoperfusion refer the patient to an advanced HF team for further management.

Gaps in evidence and research implications

Major HF clinical trials typically enrolled patients with normal BP, leaving those with very low BP (e.g. SBP < 90 or 95 mmHg) notably underrepresented.¹³² Future pragmatic trials are crucial for

addressing key research gaps in the treatment of patients in this setting. These studies should focus on:

- Evaluating the effectiveness and safety of GDMT in patients with very low BP, particularly when using medications like ACE inhibitors, ARB or ARNi that significantly lower BP.
- Testing the proposed low BP management strategies to provide randomized evidence in support of this expert consensus.
- Investigating the impact of discontinuing non-life-extending medications to enhance GDMT outcomes, focusing on overall patient survival and QoL, as well as the potential benefits of reducing diuretics to decrease HF-related events, including hospitalizations.
- Assessing the effects of interventional cardiac procedures (e.g. CRT, TAVR, and mitral/tricuspid TEER) and the restoration of sinus rhythm in patients with AF on BP, GDMT optimization and outcomes in patients with HF and low BP.

Addressing these research gaps is crucial for developing more effective treatment strategies for patients with HFrEF and low BP, potentially affecting global treatment approaches.

Table 3 Take home messages

- Asymptomatic or mildly symptomatic low BP should not be a reason for GDMT reduction or cessation
- Reduction or cessation of one or more GDMT is advisable in case of systolic BP <80 mmHg or low BP with relevant symptoms
- In chronic HF always evaluate symptoms of hypotension/orthostatic hypotension
- In acute HF always assess organ perfusion (shock or pre-shock)
- The BP threshold as an exclusion criterion in clinical trials is primarily set to prevent dropouts and maintain the trial's statistical power.
- SGLT2is and MRAs have the least effect on BP and in low BP groups they may increase BP
- In specific conditions other drugs or devices may be appropriate to facilitate optimization of GDMT (ivabradine, CRT, TEER, digoxin, ...)
- In low BP conditions, the first step is to stop unnecessary cardiac medications and then look for transient and reversible causes
- Multidisciplinary care coordination of selection of appropriate alternatives with lesser BP-lowering effect for non-cardiac medications (i.e. medication for prostate hypertrophy, antidepressants)
- When discontinuation or reduction of a therapy is needed: start with the less tolerated
- In persistent hypotension with inability to initiate or titrate GDMT refer early to HF specialist or advanced therapy programmes
- When BP improves, always consider reinitiation or rechallenge of drugs based on better tolerated first
- Strive to achieve optimal therapy
 1. Space medications in divided doses throughout the day
 2. Give selective BB or consider ivabradine
 3. If ARNI not tolerated, try to introduce and optimize ACEi/ARB

ACEi, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; ARNI, angiotensin receptor–neprilysin inhibitor; BB, beta-blocker; BP, blood pressure; CRT, cardiac resynchronization therapy; GDMT, guideline-directed medical therapy; HF, heart failure; HFrEF, heart failure with reduced ejection fraction; MRA, mineralocorticoid receptor antagonist; OMT, optimal medical therapy; SGLT2i, sodium–glucose cotransporter 2 inhibitor; TEER, transcatheter edge-to-edge repair.

Conclusion

To improve the implementation of HF guideline recommendations, it is crucial to address clinical factors that limit initiation, optimization, and continuation of GDMT, in order to improve care and outcome of patients with HFrEF (Table 3). This clinical consensus statement highlights the importance of GDMT in patients with HFrEF and low BP, defines clinically relevant hypotension, propose a pathway to identify its aetiologies and strategies to manage HF treatments in different scenarios. It also emphasizes the role of non-pharmacological interventions that can help in increasing BP and optimize GDMT. The main aim of this document is to assist healthcare providers treating patients with HF (cardiologists, intensivists, and general practitioners) to overcome clinical inertia by improving the initiation and optimization of GDMT and concurrently reducing the rate of HF treatment discontinuation. Moreover, it stresses the need for physicians in other subspecialties to consult with the HF team before reducing or withholding HFrEF medications due to the perception of 'low BP' readings. Finally, it issues a call to action to address the significant knowledge gaps in this area.

Supplementary Information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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